Error Analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One error-analysis and next-test record.

Trace slice rates,compare versions and justify a controlled follow-up with honest limits.

Keep one habit: W1–W3/W5 → selected route comparison → project error log → fresh changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Nearest Neighbors. Give an example and a mistake that the idea helps you avoid.

Precision and Recall

My remembered idea, example and correction

Confusion Matrix

My remembered idea, example and correction

Nearest Neighbors

My remembered idea, example and correction

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main fictional development replay: dim12 cases with old4/new2 errors; bright18 with old3/new4. Rates dim33.3%→16.7%, bright16.7%→22.2%; total7/30→6/30. D01–D04 old mistakes, D01–D02 new; B01–B03 old, B01–B04 new. Lighting/device/route co-vary, so no causal lighting effect is isolated. Separate picture bright1/8=12.5%,dim3/6=50%,total4/14;dim replay1/6 is reused development evidence.

  Follow-up new scene pairs S1–S4 under both lights is proposed, with no measured outcomes supplied.

- Reference for this exercise: open this week's student lesson, "Error Analysis", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Calculate the original error rates by lighting condition.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main fictional development replay: dim12 cases with old4/new2 errors; bright18 with old3/new4. Rates dim33.3%→16.7%, bright16.7%→22.2%; total7/30→6/30. D01–D04 old mistakes, D01–D02 new; B01–B03 old, B01–B04 new. Lighting/device/route co-vary, so no causal lighting effect is isolated. Separate picture bright1/8=12.5%,dim3/6=50%,total4/14;dim replay1/6 is reused development evidence.

  Follow-up new scene pairs S1–S4 under both lights is proposed, with no measured outcomes supplied.

- Does the difference prove dim light causes model errors?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main fictional development replay: dim12 cases with old4/new2 errors; bright18 with old3/new4. Rates dim33.3%→16.7%, bright16.7%→22.2%; total7/30→6/30. D01–D04 old mistakes, D01–D02 new; B01–B03 old, B01–B04 new. Lighting/device/route co-vary, so no causal lighting effect is isolated. Separate picture bright1/8=12.5%,dim3/6=50%,total4/14;dim replay1/6 is reused development evidence.

  Follow-up new scene pairs S1–S4 under both lights is proposed, with no measured outcomes supplied.

- Summarize the changed results without hiding a **regression**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main fictional development replay: dim12 cases with old4/new2 errors; bright18 with old3/new4. Rates dim33.3%→16.7%, bright16.7%→22.2%; total7/30→6/30. D01–D04 old mistakes, D01–D02 new; B01–B03 old, B01–B04 new. Lighting/device/route co-vary, so no causal lighting effect is isolated. Separate picture bright1/8=12.5%,dim3/6=50%,total4/14;dim replay1/6 is reused development evidence.

  Follow-up new scene pairs S1–S4 under both lights is proposed, with no measured outcomes supplied.

- A team removes all disputed records only from the new model report. Why is the comparison invalid?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- This selected route task uses its own group sizes and old/new error counts, not the main lighting or pictured cohort. Keep A's denominator10 and B's20 for both methods. The two route groups partition this supplied30-case comparison; no cause, individual IDs or actual change implementation is provided. Report both aggregate and group directions before recommending a next test.

- Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create the Plant Classifier error log linking case ID, actual, predicted, confidence, condition, likely cause, and next action.

- The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Explain the limit for **Error analysis** and controlled follow-up. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Error analysis** and controlled follow-up. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Four errors occur among five backlit photos; two among twenty ordinary photos.
- Compare rates and investigate.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Of eight errors, six are on blurry images and two on clear images. Can you conclude blurry images have a higher error rate without group totals?
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 18 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A model fails 6/10 backlit and 2/20 front-lit photos. Compute error rates and design a follow-up that tests lighting rather than leaf identity.

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Old errors are 5/10 on route A and 2/20 on B. New errors are 3/10 and 5/20. State total and group changes and a justified next step.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

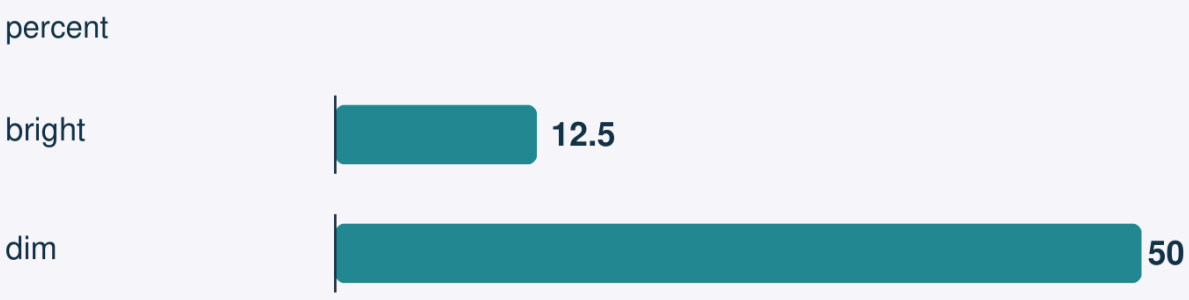
Task and evidence record

- Find an error cluster before choosing a controlled follow-up.
- Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/18

Errors per 100 photos in each lighting slice



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/18

Error Analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice errors

Bright: 12.5%; dim: 50%. Total errors4/14, about28.6%.



2. Inspect dim failures

Check blur, lighting and label uncertainty instead of assuming lighting caused every error.



3. Plan controlled pair

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

CHANGE ONE THING

Dim errors fall to1/6 in a later same-case replay.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Find an error cluster before choosing a controlled follow-up.
- Which labelled step matches this detail? Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.
- 1. Plan controlled pair
- 2. Compute slice errors
- 3. Inspect dim failures
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Plan controlled pair Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Read the picture in order.

Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Dim errors fall to1/6 in a later same-case replay. Which conclusion is supported by the supplied facts?
- 1. The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision.

Observed improvement on reused cases is not an untouched final generalisation test.

Repair and keep the record

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Nearest Neighbors (ai-4/12). Goal: Compute absolute distances and rank **training** neighbours. Short practice: Why is the k=1 outcome at length5 not uniquely determined without a tie rule?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

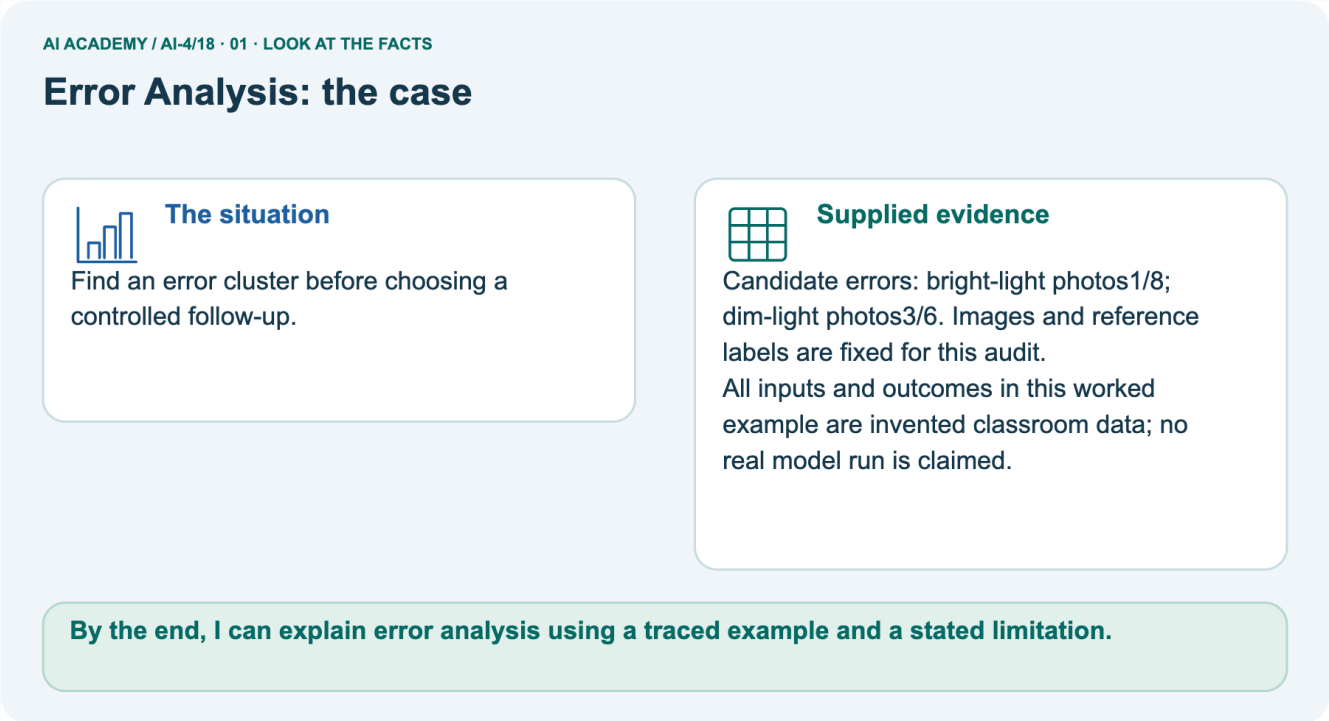
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the relevant picture instead of repeating every task.

01. Error Analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

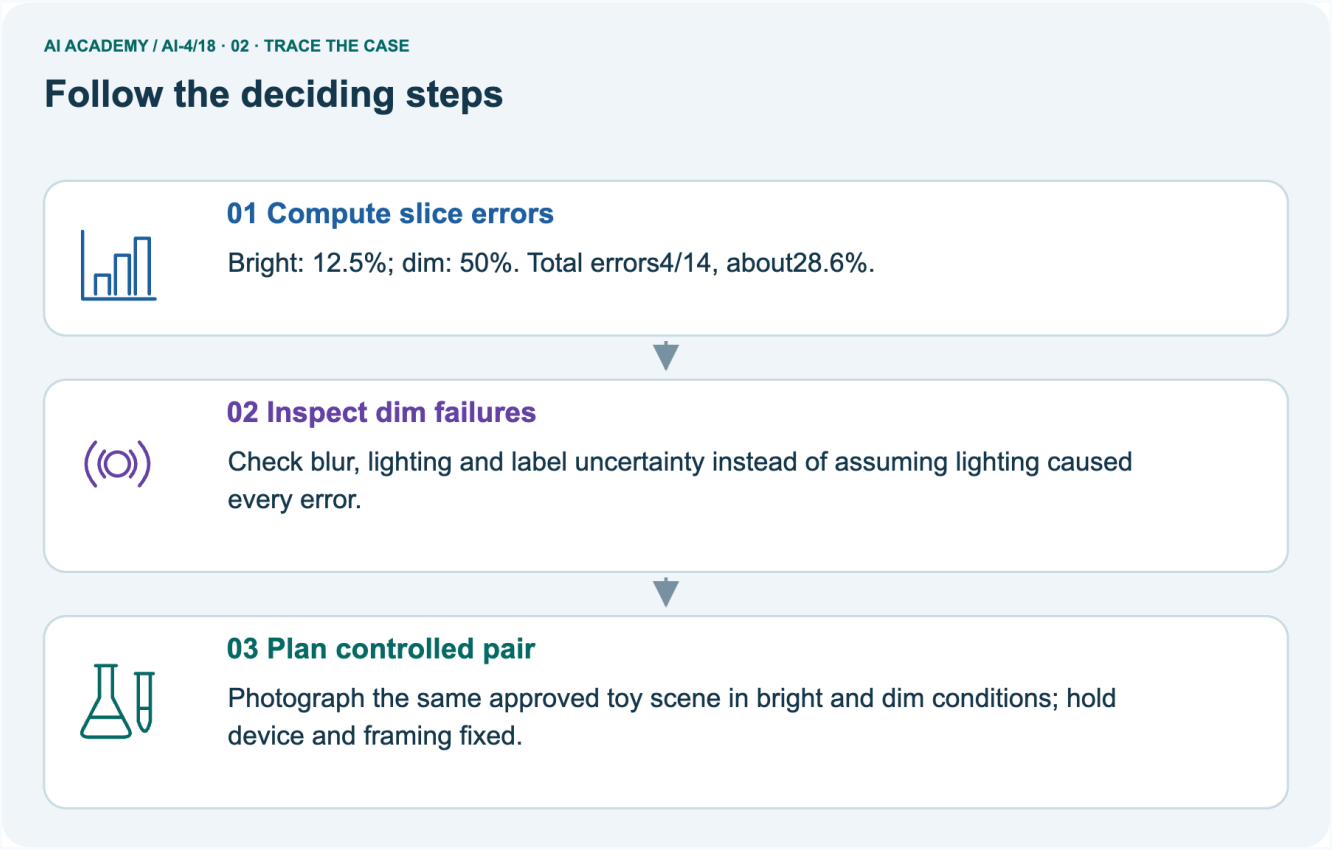
Find an error cluster before choosing a controlled follow-up. Supplied facts: Candidate errors: bright-light photos1/8; dim-light photos3/6. Images and reference labels are fixed for this audit.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

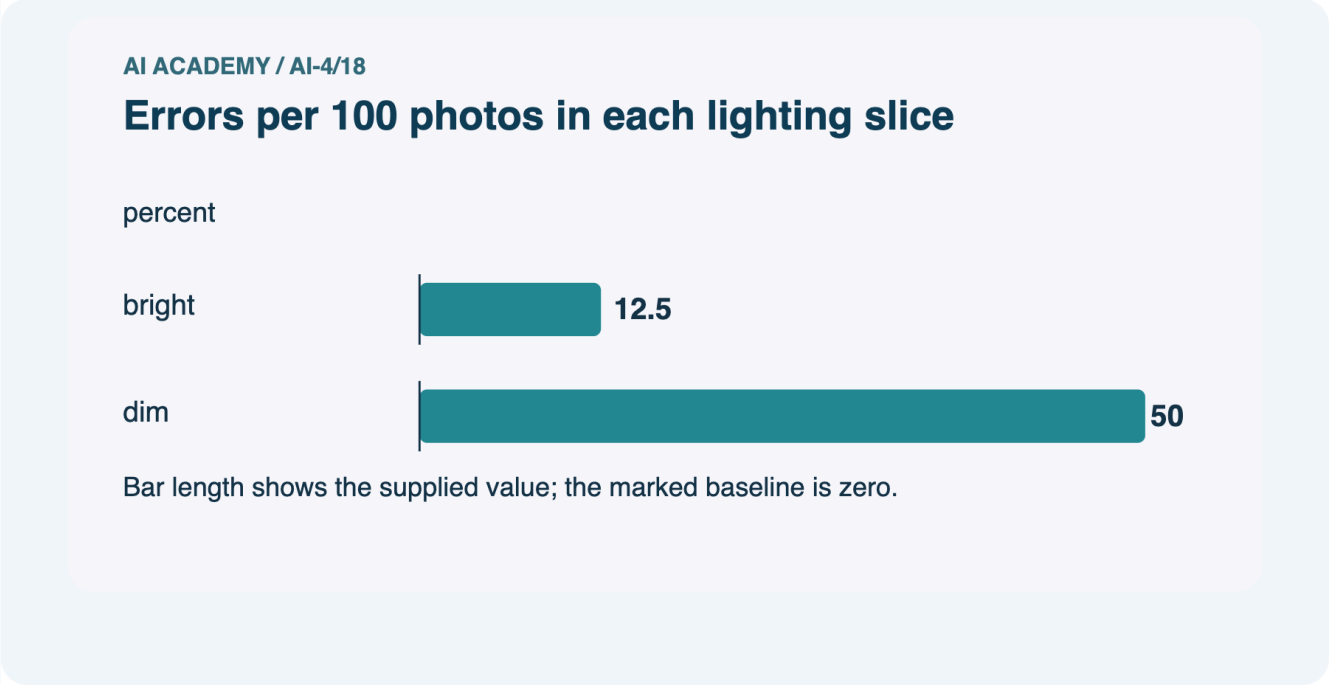
Compute slice errors: Bright: 12.5%; dim: 50%.

Total errors4/14, about28.6%. → Inspect dim failures: Check blur, lighting and label uncertainty instead of assuming lighting caused every error. → Plan controlled pair: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Errors per 100 photos in each lighting slice



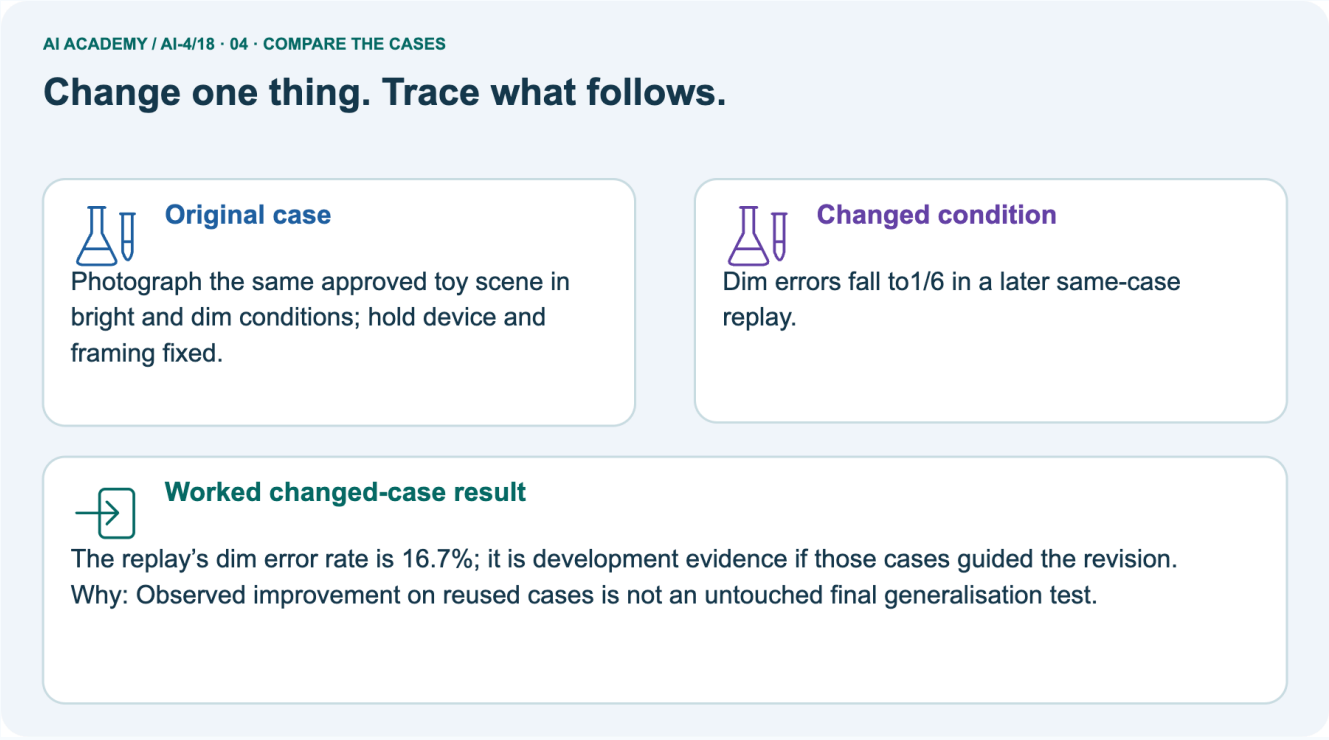
Follow the labelled picture with the matching explanation. Open the vector diagram

Errors per 100 photos in each lighting slice. Units: percent. bright: 12.5. dim: 50.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

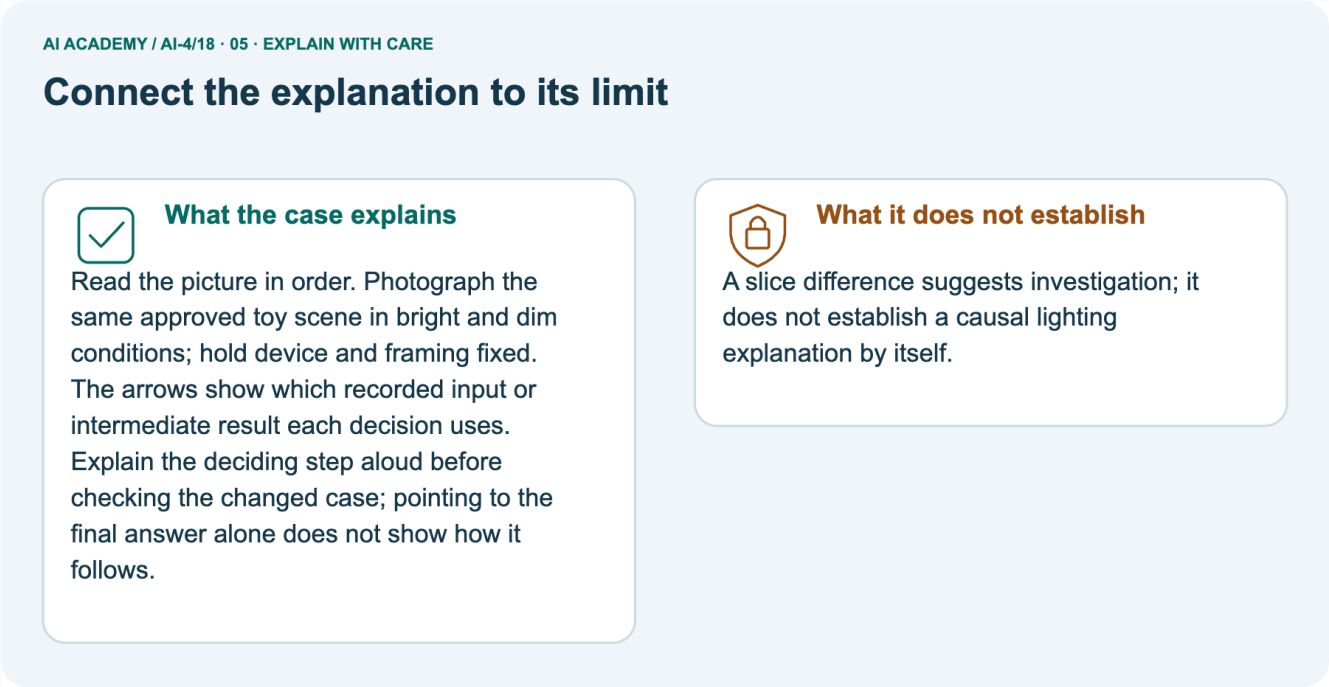
Original case: Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. Changed condition: Dim errors fall to1/6 in a later same-case replay.

Expected result: The replay's dim error rate is 16.7%; it is development **evidence** if those cases guided the revision. Reason: Observed improvement on reused cases is not an untouched final generalisation test.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Photograph the same approved toy scene in bright and dim conditions; hold device and framing fixed. The arrows show which recorded **input** or intermediate result each decision uses.

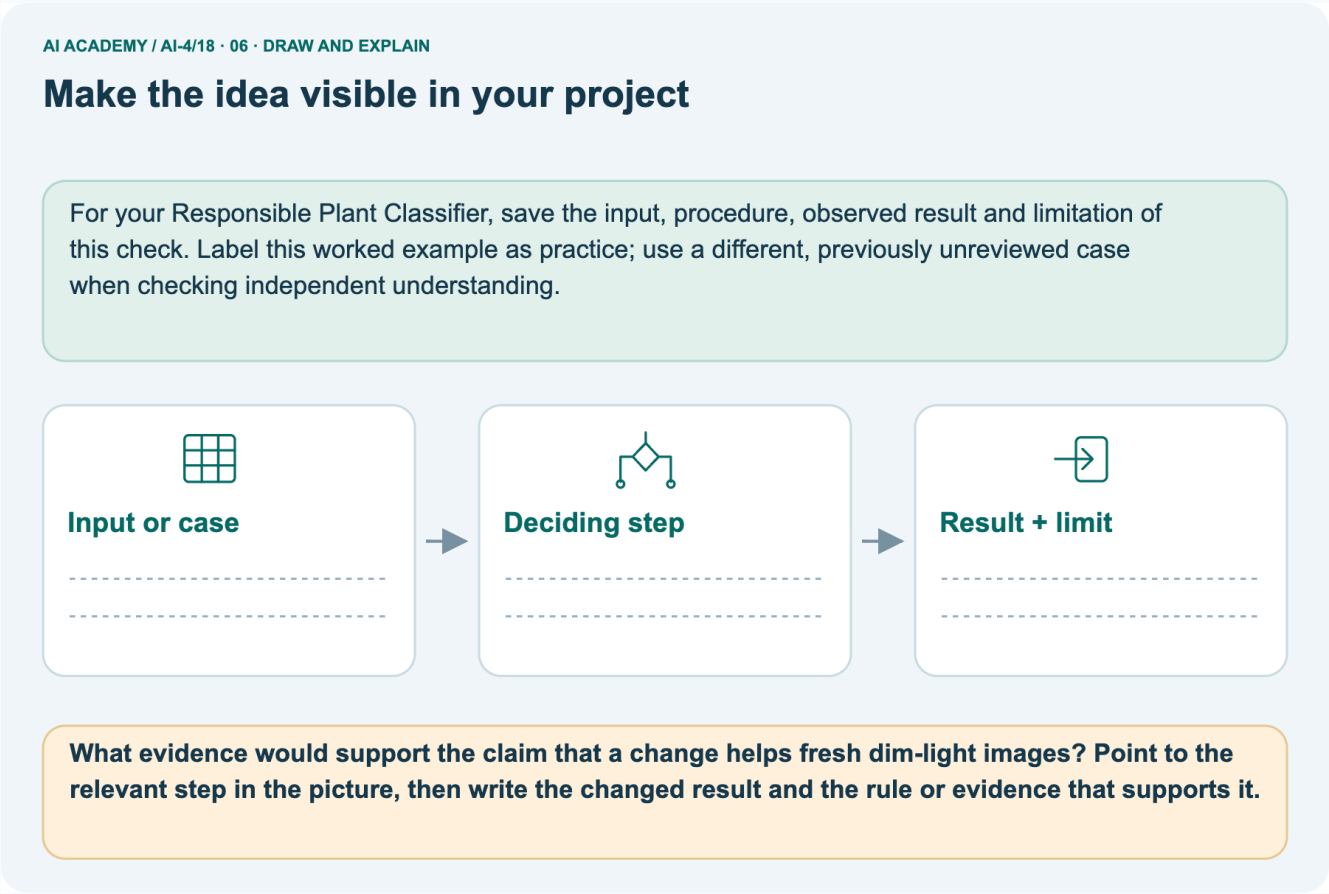
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A slice difference suggests investigation; it does not establish a causal lighting explanation by itself.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** would support the claim that a change helps fresh dim-light images? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Error Analysis: the case

02. Follow the deciding steps

03. Errors per 100 photos in each lighting slice

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.